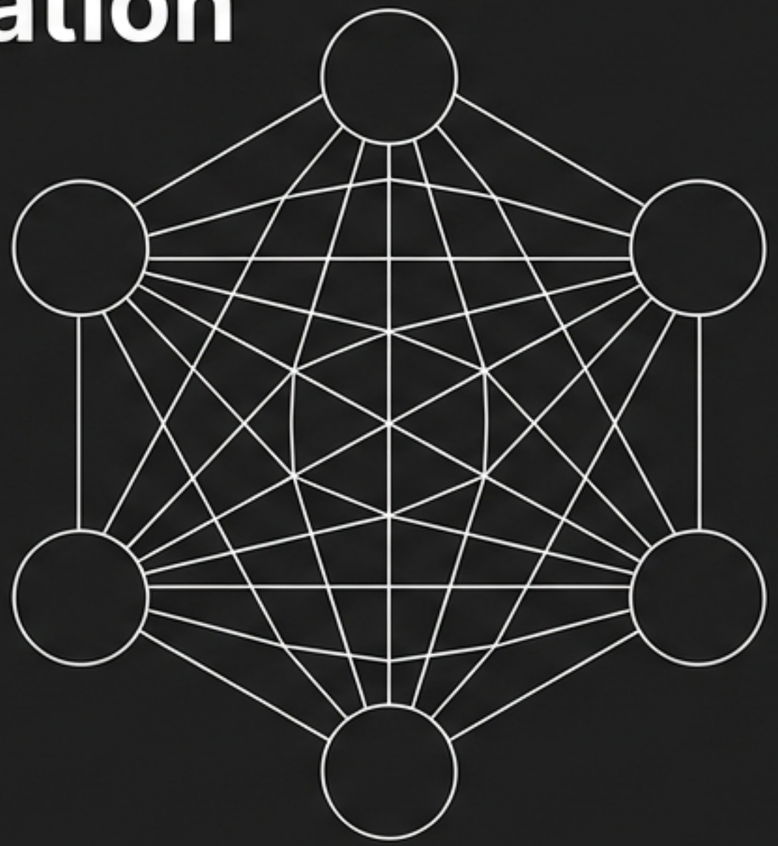
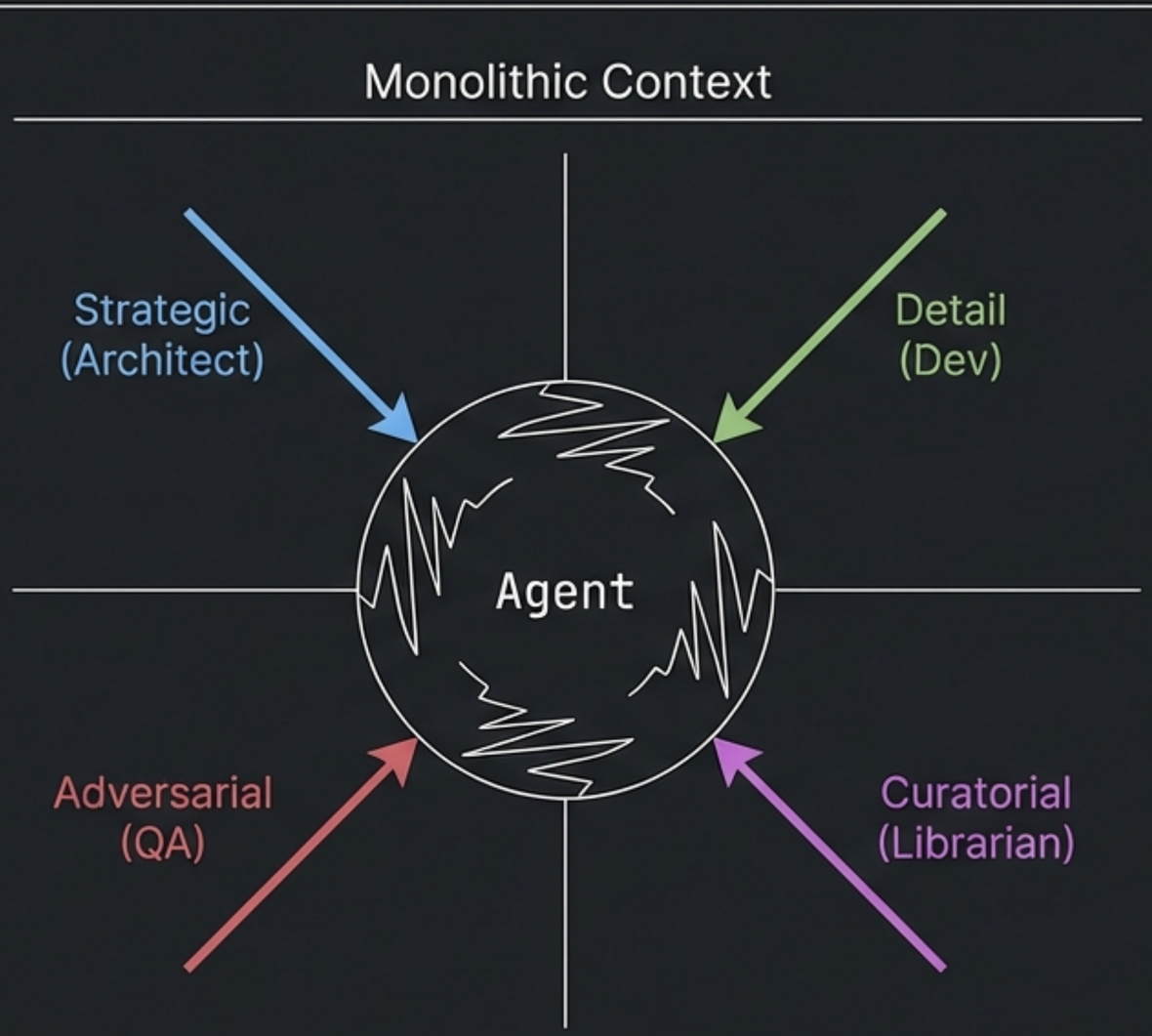
Issues-FS: Role-Based Agent Coordination

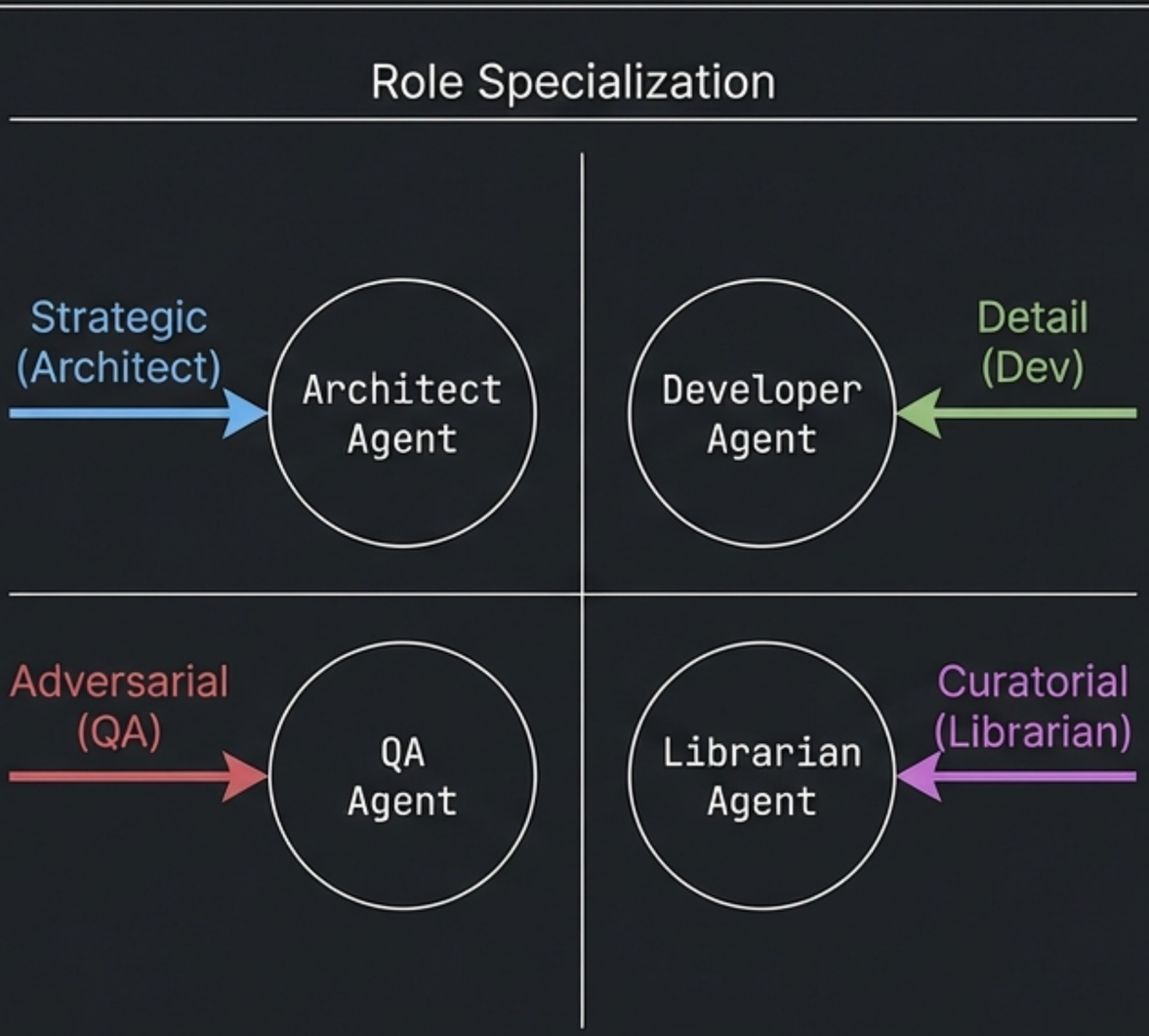
A submodule architecture for specialized, self-managing AI teams.



Version: v1.0
Date: 05 Feb 2026
Status: Draft
Dependency: Issues-FS__Architecture-Overview v1.0

The Problem: Context Pollution





Core Argument: When a single agent attempts to wear multiple hats—writing code, reviewing it, and managing releases—quality degrades. Team members are instances of the same model, differentiated only by context.

The Solution: The 'Role Repo' Pattern

```
Issues-FS__Dev/
├── modules/              # Source Code
│   ├── ...
├── roles/                # Agent Definitions
│   ├── Issues-FS__Dev__Role__Conductor/
│   │   └── ...
│   ├── Issues-FS__Dev__Role__Dev/
│   │   └── ...
│   ├── Issues-FS__Dev__Role__QA/
│   │   └── ...
│   └── Issues-FS__Dev__Role__Architect/
│       └── ...
└── ...
```

The Insight: The submodule pattern works for agent roles just as well as it works for code.

A **"Role Repo"** is a repository where the primary artifact is agent configuration, not just source code.

The Ensemble: Six Foundational Roles

Conductor

Orchestration, Blockers, Sprint Planning.

Architect

Strategy, API Design, ADRs.

Dev

Implementation, Source Code, Unit Tests.

QA

Quality Gates, Integration Tests, Defects.

DevOps

CI/CD, Infrastructure, Releases.

Librarian

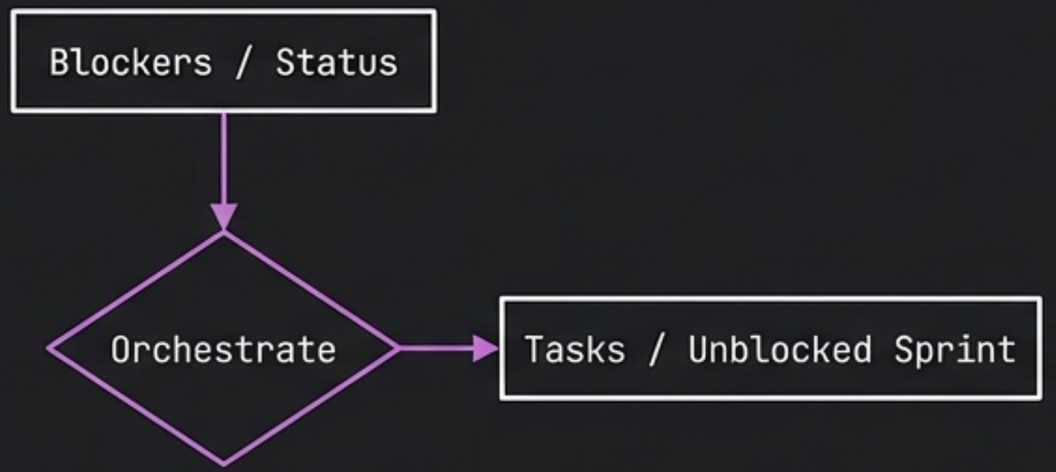
Documentation, Curation, "Truth".

These roles cover the core development workflow. Each has a clear, non-overlapping scope.

Strategic Roles: Conductor & Architect

The Conductor

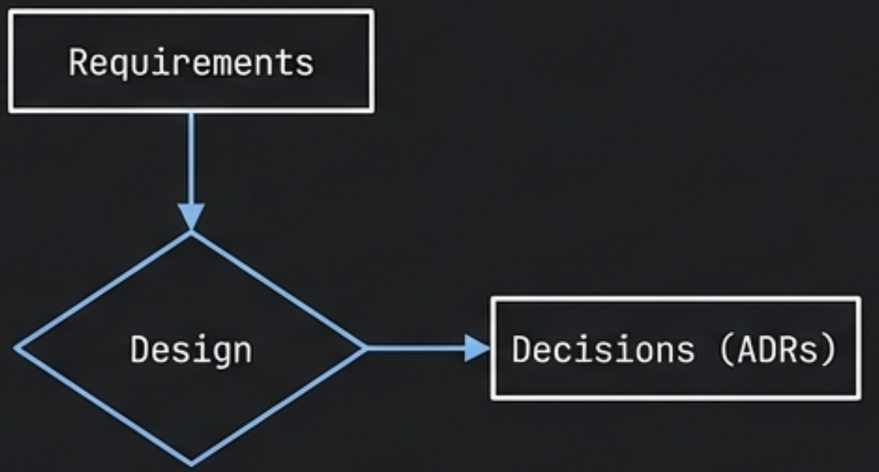
Operational Management



- Owns the Sprint
- Hands off to: All Roles
- Distinction: Not a Project Manager. Orchestrates without doing the work.

The Architect

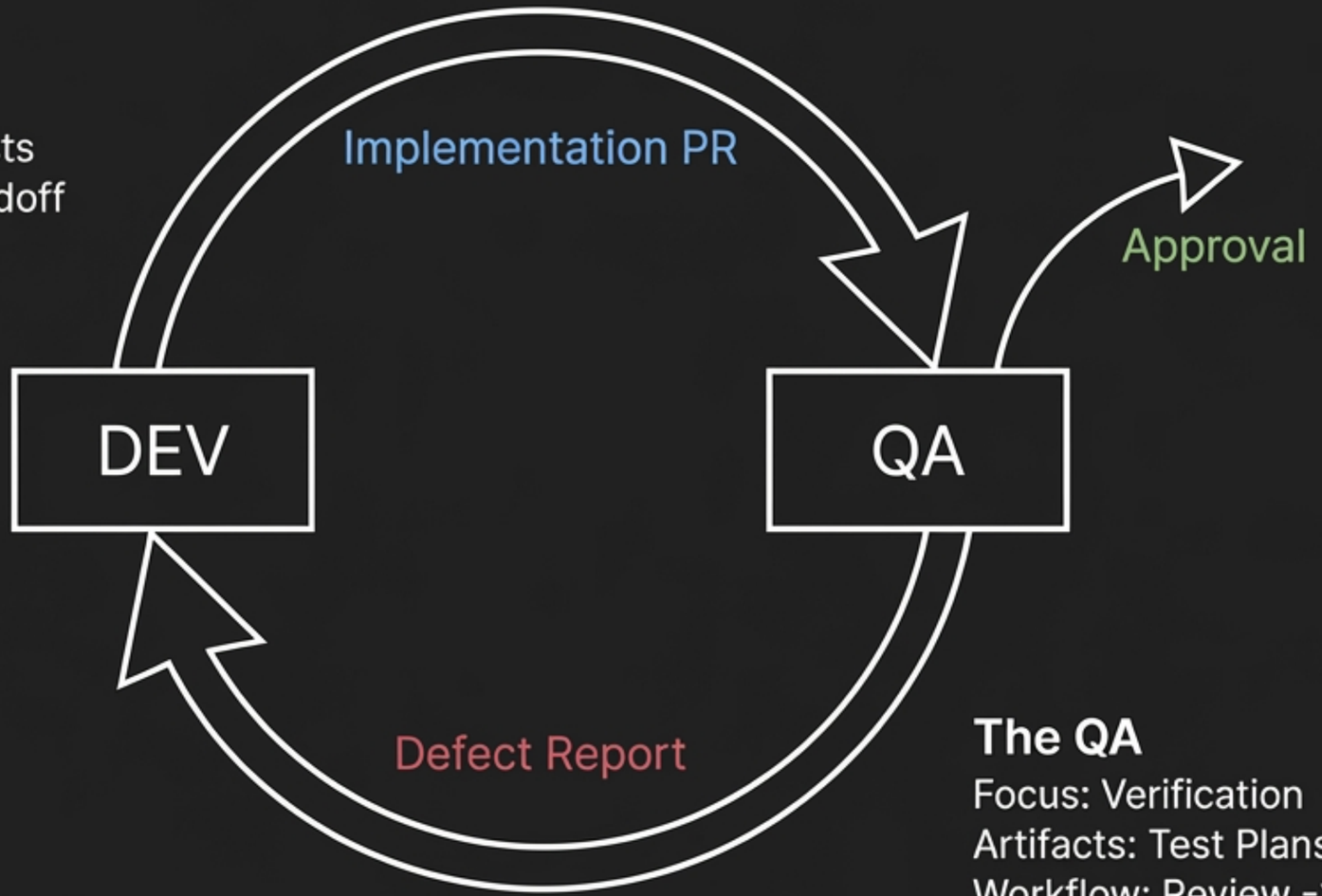
Technical Strategy



- Owns the Decision
- Hands off to: Dev, Librarian
- Key Artifact: Architecture Decision Records

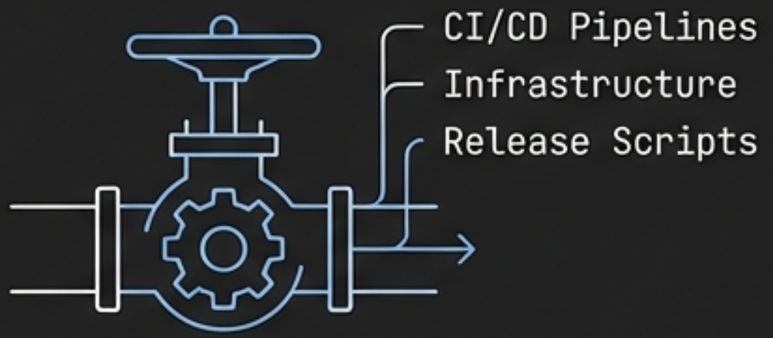
Execution Roles: Dev & QA

The Dev
Focus: Implementation
Artifacts: Source Code, Unit Tests
Workflow: Task -> Code -> Handoff



The QA
Focus: Verification
Artifacts: Test Plans, Integration Tests
Workflow: Review -> Approve OR Defect

Support Roles: DevOps & Librarian



RELEASE PIPELINE

DevOps: The Gatekeeper

Responsibilities: CI/CD Pipelines, Infrastructure, Release Scripts.

Authority: The only role that can trigger a Release issue. Protects production from bad code.

KNOWLEDGE REPOSITORY

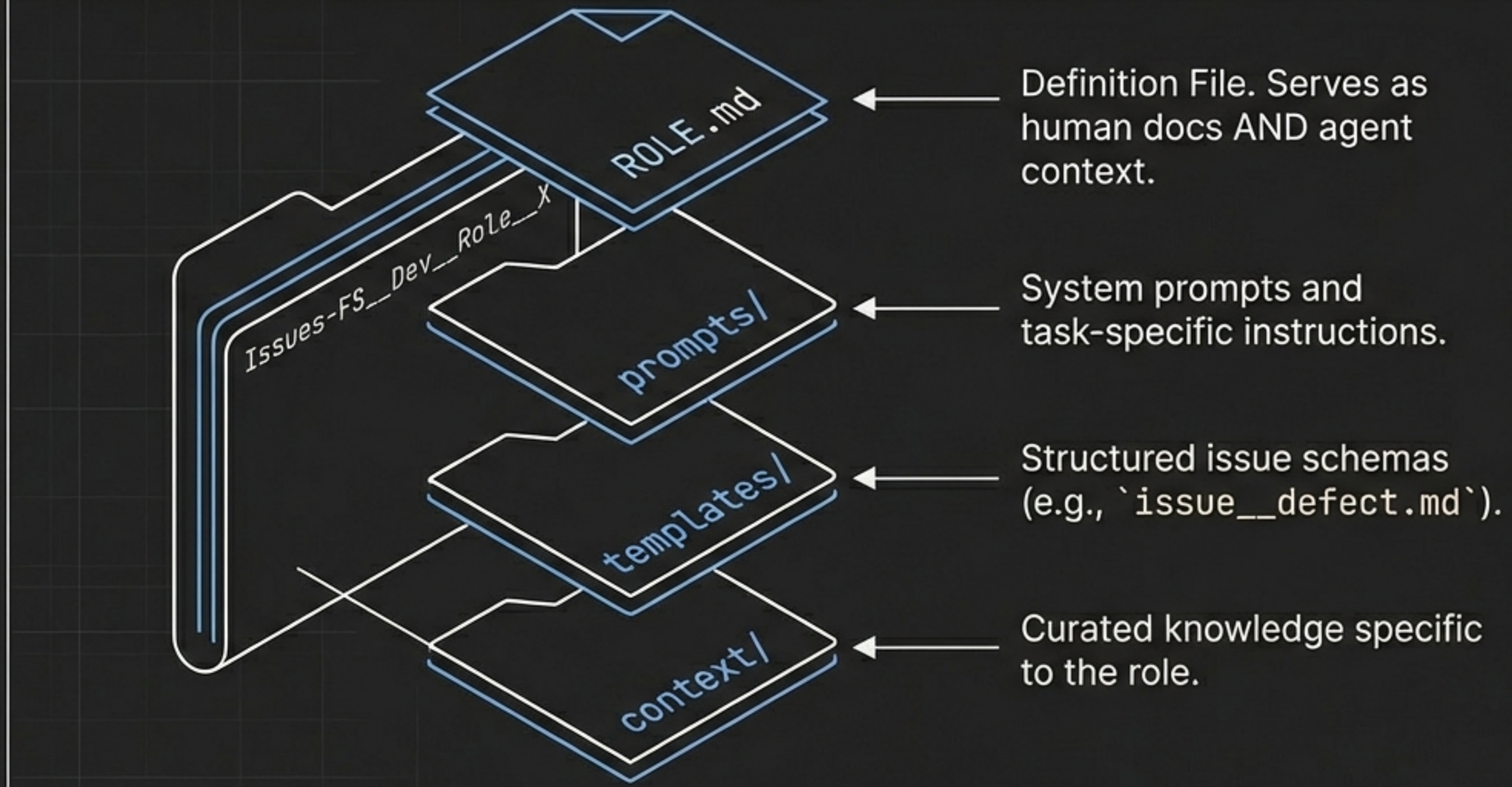
Librarian: The Curator

Responsibilities: Knowledge Coherence, Changelogs, Onboarding.

Philosophy: "Docs" implies a static dump. "Librarian" implies judgment—deciding what is canonical and what is stale.

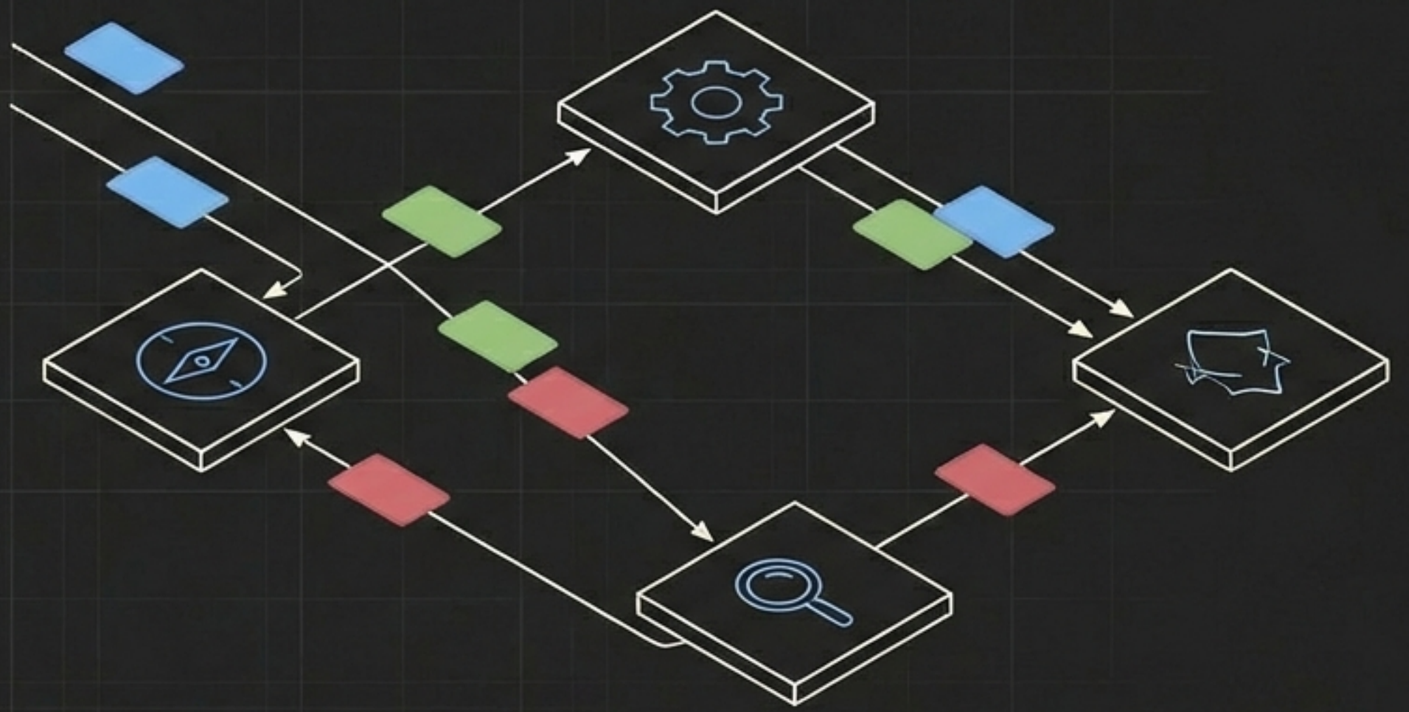
Anatomy of a Role Repo

A role repo encodes responsibilities, boundaries, and coordination protocols.



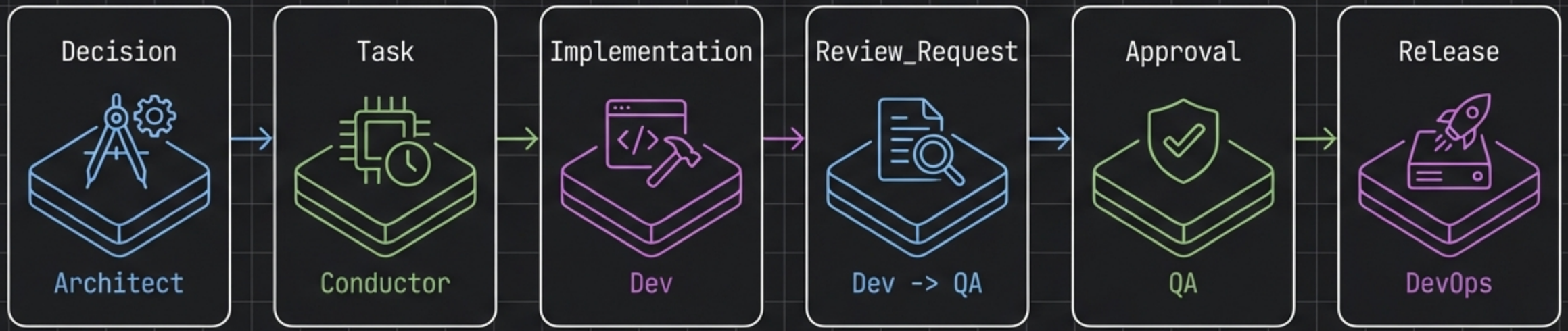
Coordination via Typed Issues

1. **Decision** (Architect)
   -> Strategic Choice
2. **Handoff** (Any)
   -> Transfer of Work
3. **Review_Request** (Any)
   -> Asking for Approval
4. **Defect** (QA)
   -> Bug Report
5. **Blocker** (Any)
   -> Escalation
6. **Release** (DevOps)
   -> Deployment Candidate



"Issues are the '**State Machine**'. Communication happens via **structured data**, not chat logs."

The Workflow State Machine



Traceability: The entire flow is tracked as a graph query.
"Show me all issues linked to this Decision".

Dogfooding as Architecture

Concept:

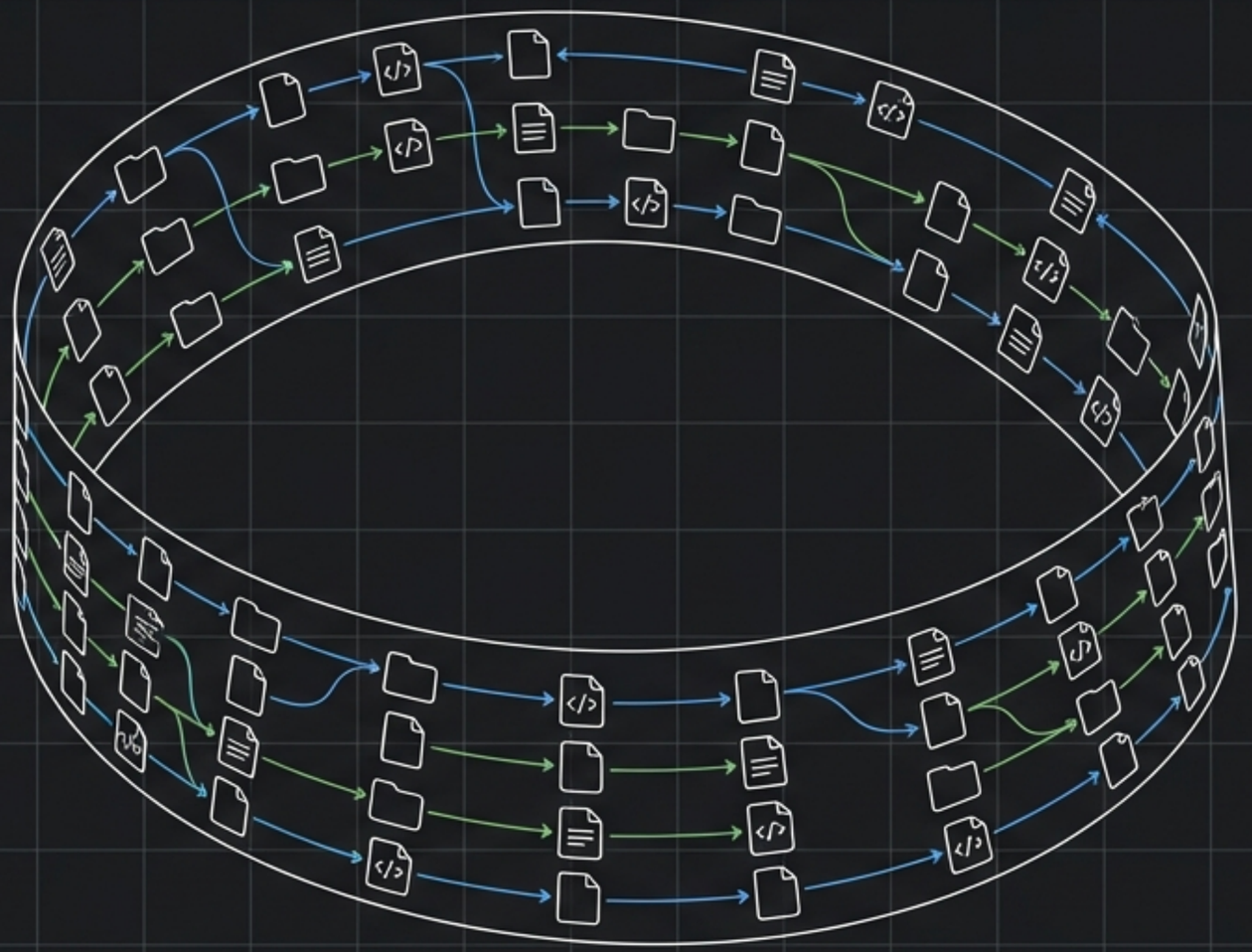
`Issues-FS` manages itself using `Issues-FS`.

Mechanism:

- Each `role repo` tracks its own work as `Issues-FS issues`.
- Cross-role coordination uses the `Issues-FS graph`.

Impact:

Improvements to the platform immediately improve the process of building the platform.

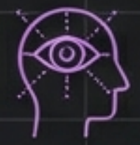


Scaling the Ecosystem

Future Roles

Security: Threat modeling, auditing

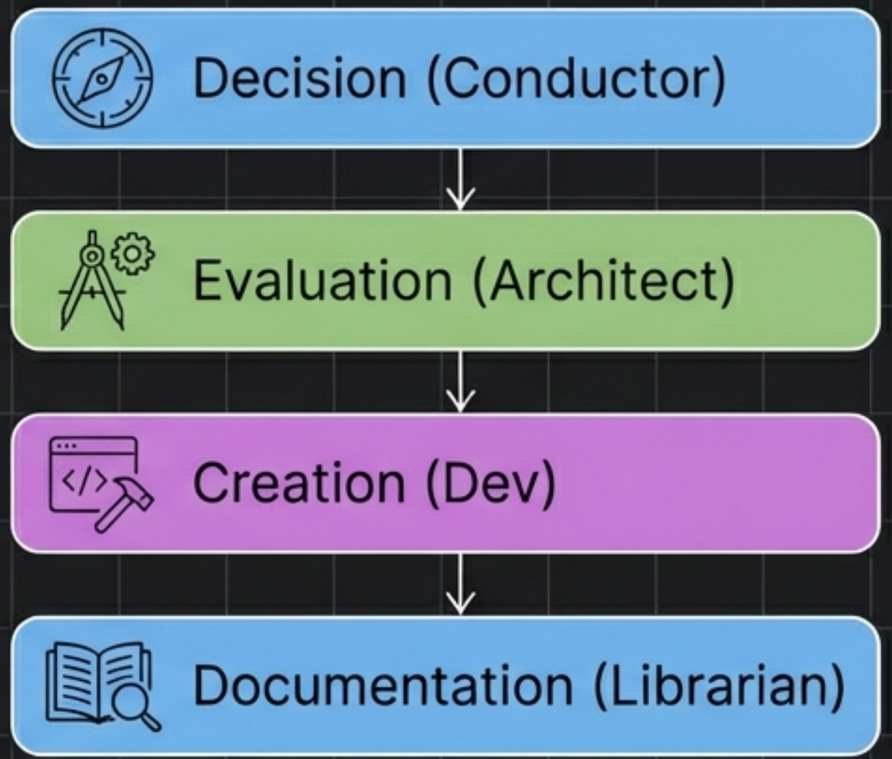
Performance: Benchmarking, SLAs



UX: User research, Design

Data: Schema evolution

Role Addition Protocol



`Guiding Principle: Create highly focused teams with just the resources needed.`

Key Architectural Decisions (ADRs)

D1: Role vs. Persona	Roles define ownership. Personas define tone. In coordination, ownership matters more.
D2: Conductor vs. PM	“Project Manager” carries baggage. “Conductor” implies orchestration of an ensemble without doing the grunt work.
D4: Decisions are Issues	Every decision gets a unique ID and history. This enables graph queries on the decision tree itself.
D5: Typed Issues	The issue schema IS the workflow engine. No external orchestration tool is needed.

Summary & Status

```
Document:       issues-fs__role-based-agent-coordination
Version:        v1.0 (Draft)
Date:           2026-02-05

Dependencies:
 - Issues-FS__Architecture-Overview
 - Memory-FS
 - MGraph-DB
 - OSBot-Utils
```

The submodule pattern works for code. The insight is that it works equally well for agent roles.